

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India **09/2021 – 01/2025 (3.5 Years)**

- Analyzed structured and unstructured data from semiconductor fabrication processes, delivering insights into process health and enabling data-driven decisions for improvement activities, achieving a **15% reduction** in yield excursions.
- Collaborated with Technology Development and Local Yield teams to import new semiconductor technology into production fabs, ensuring successful product introduction and optimization for yield and performance targets.
- Established and implemented workflows and procedures for continuous improvement programs, ensuring quality and preventing yield excursions in line and sort processes by integrating **Python** automation scripts.
- Assessed quality and reliability risks for manufacturing materials and process changes, ensuring adherence to Intel's standards by developing comprehensive risk assessment reports and cross-functional review processes.
- Owned product maintenance for semiconductor manufacturing lines, ensuring products ran on target and optimizing targets for yield, device, and performance through iterative process adjustments and data analysis.
- Managed production support and incident response for enterprise applications using **Splunk** and **Dynatrace**, resolving P1/P2 incidents within strict SLA targets and maintaining **99.9%** system uptime.

SKILLS

Semiconductor Manufacturing – FEOL/BEOL Integration, Yield Analysis, Process Optimization, Lithography, Defect Reduction, Continuous Improvement, Quality Risk Assessment.

Data Analysis & BI – Structured and Unstructured Data Analysis, Statistical Analysis, Power BI, Tableau, DAX, Power Query, Data Modelling.

Programming & Scripting – Python, Java, Spring Boot, SQL (PostgreSQL, MySQL, Oracle), Shell Scripting.

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Docker, Linux.

DevOps & CI/CD – Jenkins, Git, Jira, Confluence.

Observability & Monitoring – Splunk, Dynatrace, CloudWatch.

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin **02/2026**

BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India **01/2021**

PROJECTS

Semiconductor Process Yield Optimization Platform (HCL Technologies)

- Developed a data analytics platform using **Python** and **Pandas** to process structured and unstructured data from fabrication lines, identifying root causes of yield loss.

- Integrated real-time monitoring dashboards in **Power BI**, visualizing key process parameters and enabling proactive identification of potential yield excursions, reducing critical incident response time by **30%**.
- Implemented automated reporting workflows using **SQL** queries against PostgreSQL, delivering weekly yield analysis reports to cross-functional teams, improving decision-making speed by **25%**.

Financial Services Transaction Monitoring Dashboard (HCL Technologies)

- Enhanced a payments operations dashboard by implementing new transaction monitoring features using **Java Spring Boot**, increasing visibility into payment message flows.
- Automated data validation processes with **Python** scripting, reducing manual data reconciliation efforts by **40%** and improving data accuracy for audit trails.
- Managed incident response and production support for the dashboard, resolving critical issues within defined SLAs and ensuring **99.95%** uptime for operational alerting.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)